

Statistical modelling and simulation

The course provides a foundation for problem solving in technology, economy and science using **statistical modeling, simulation and analysis**.

A key motivation for the course is that there are numerous situations which are far too complicated for exact calculations, but where an (approximate) answer (easily) can be obtained by proper statistical modelling and simulations.

In this course you will learn:

- Key techniques for statistical modelling and simulation.
- To make and use statistical models for a wide range of problems.
- How to implement the models using R.
- Carry out simulations of statistical models, analyze the result statistically and present the result.
- Make assessments of uncertainty in the results.

Content

Part I. Foundations

- Learning to use R.
- Repetition of basic topics like:
 - Random variables
 - Discrete and continuous probability distributions
 - CDF, joint and conditional distributions
 - Expectation, variance, covariance
 - Specific probability distributions.

Part II. Estimation and Monte Carlo Foundations

- Summary statistics, estimation, uncertainty assessment.
- Useful limit laws (LLN and CLT).
- ECDF, histograms, KDE

Part III. Simulation

- Methods for generating random variables: Inversion, acceptance-rejection, transformation.
- Sums and mixtures, multivariate normals
- Simulating stochastic processes: Counting processes (incl. Poisson), Markov.

Part IV. Monte Carlo Methods

- Monte Carlo integration.
- Variance reduction.
- Bootstrapping.

Part V. Bayesian Computation

- Bayesian statistics
- Markov chain Monte Carlo (RWMH, MH, MALA, Gibbs).

Implementing all the above methods in R.

Assumed background knowledge:

- An introductory course in probability and statistics.
- Standard basic courses in calculus.
- Some basic knowledge about programming.

Introductory example I

There are numerous probability problems in everyday life which are very hard to calculate, but easy to simulate. Here is one such example.

The birthday problem

- In a group of n persons, what is the probability that at least k of these have birthday on the same day?

For $k = 2$ this is relatively easy to calculate exactly, for $k > 2$ very hard.

With simulations the problem is easy for any k .

How to simulate the birthday problem

1. Draw randomly a number from the uniform distribution on the integers $1, 2, \dots, 365$. Repeat this n times.
2. Check the n numbers and see if there is some number(s) which is drawn k or more times.
3. Repeat 1. and 2. a large number of times.
4. Calculate the proportion of times when there is some number which is drawn k or more times.

See the R-file `birthday_problem.R`.

Introductory example II

A typical real life problem contains several sources of uncertainty that need to be combined.

A Covid19 model

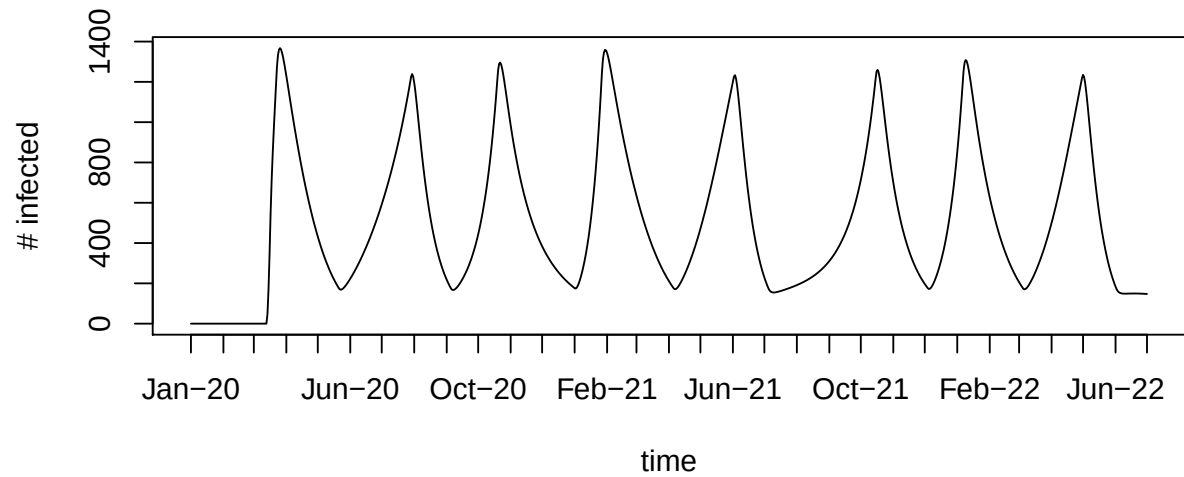
An example is a real life disease model for Covid19, roughly similar to Kissler et al (2020). The model is based on a set of 12 coupled ordinary differential equations.

The model takes into account effects of social distancing, which is installed if the infection rates passes some threshold.

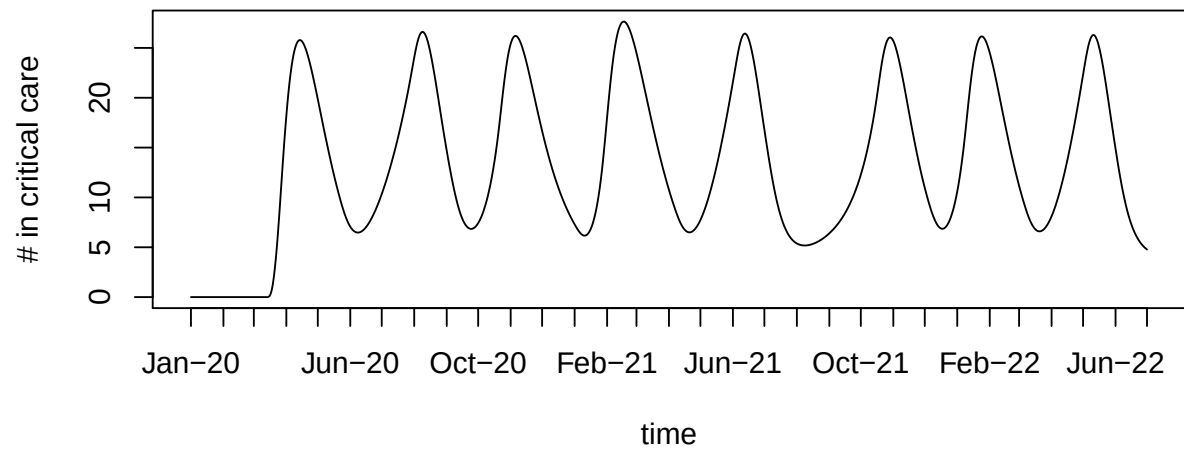
Disclaimer; several of the parameters used in the model are "made up", do not use this model for real life predictions!

Example of model output

number of Covid19 cases in Stavanger region



number of Covid19 cases in critical care at Stavanger hospital



A typical real life problem contains several sources of uncertainty that need to be combined.

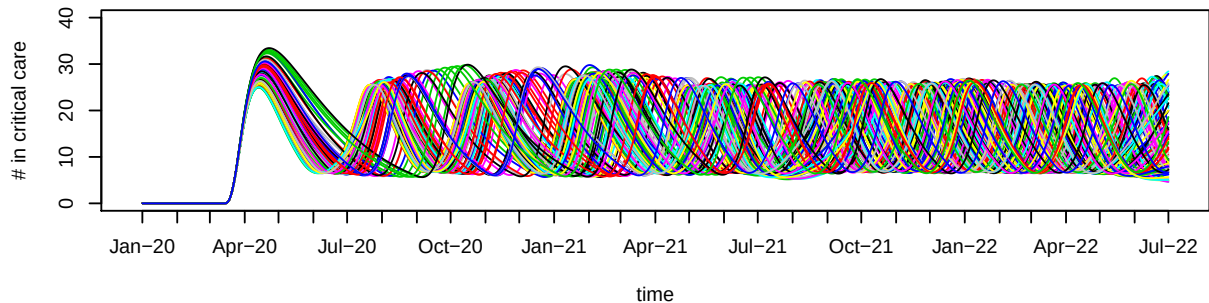
Let's consider the case where both the maximum $R_0 \in (2, 2.4)$ (i.e. number of people each infected infects) and the seasonal sensitivity of infection f (i.e. lower in summer, similar to Influenza) are uncertain.

In reality, there are many more uncertainties, but let's consider these as a start.

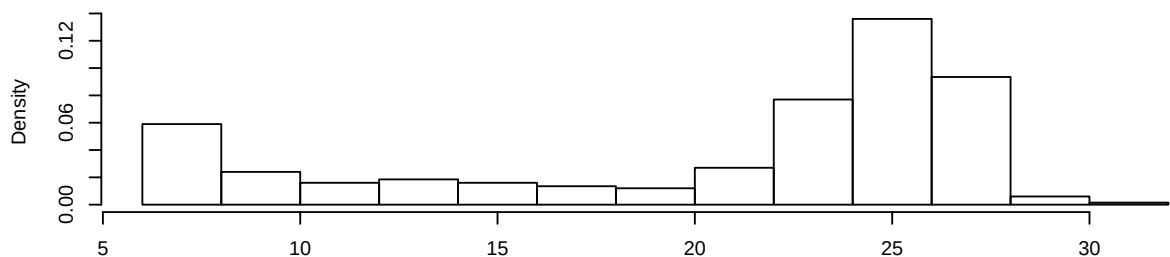
Overarching idea: simulate many "histories" for random values of R_0 and f , and calculate the probabilities you are interested in based on these realizations:

Number of patients in critical care

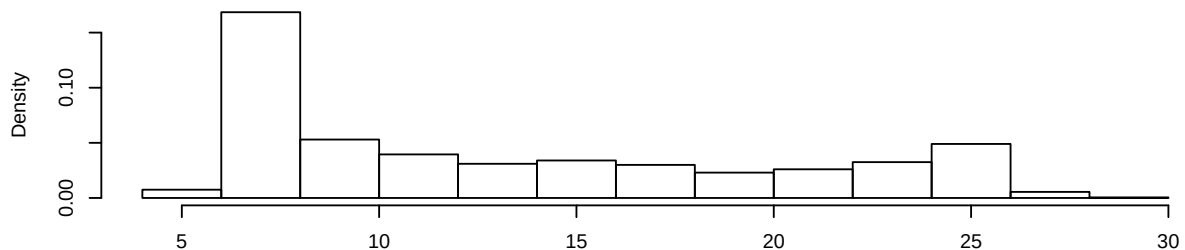
number of Covid19 cases in critical care at Stavanger hospital



number in critical care on Nov 01 2020



number in critical care on July 01 2021



Could also e.g. look at the distribution of max # patients in critical care over some time interval. Notice, essentially impossible to do analytical calculations, but really easy to simulate.

Introductory example III

Segmentation of objects from medical images of low quality or with missing data.

Shape analysis

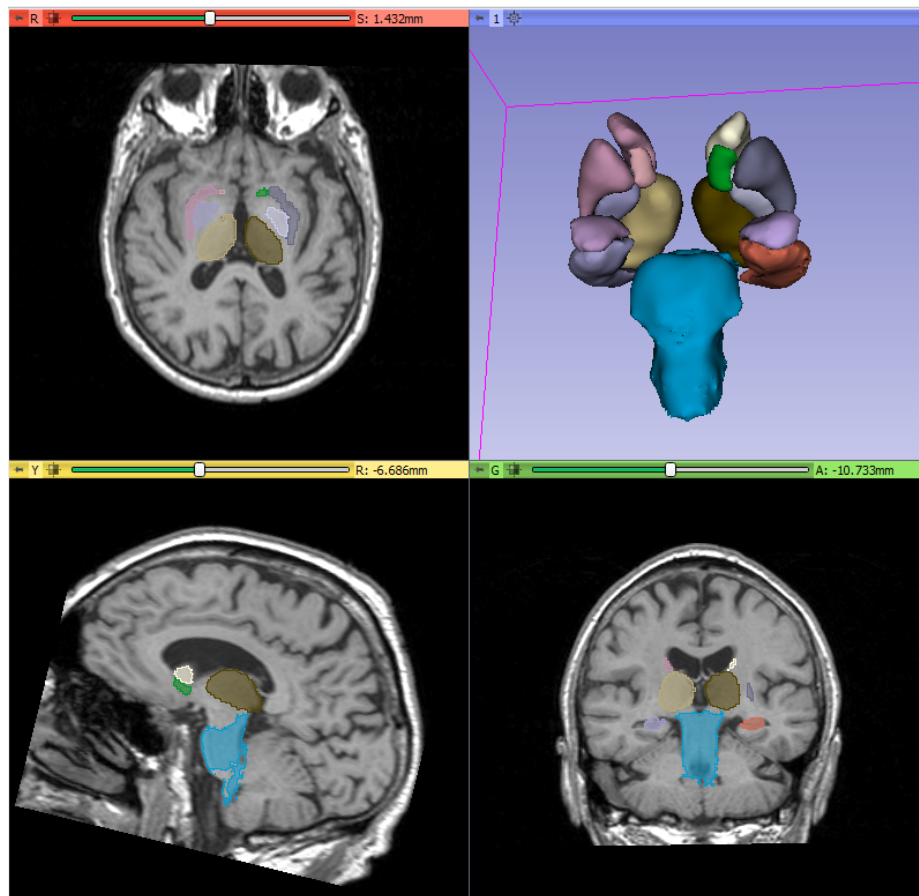
- Defining a shape model for 3-dimensional objects.
- Generating a shape distribution from training data.
- Using the shape distribution as an prior information in a Markov chain Monte Carlo (MCMC) based image segmentation.

Prostate in CT

Low image contrast between the prostate and adjacent soft tissue



Brain objects (ParkWest)



Skeletal models

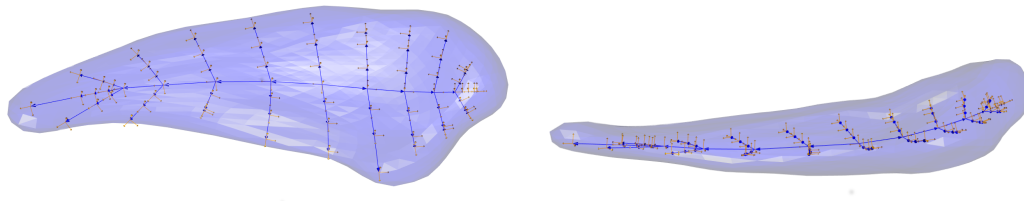


Figure 1: Fit of skeletal model to a caudate (two different views)

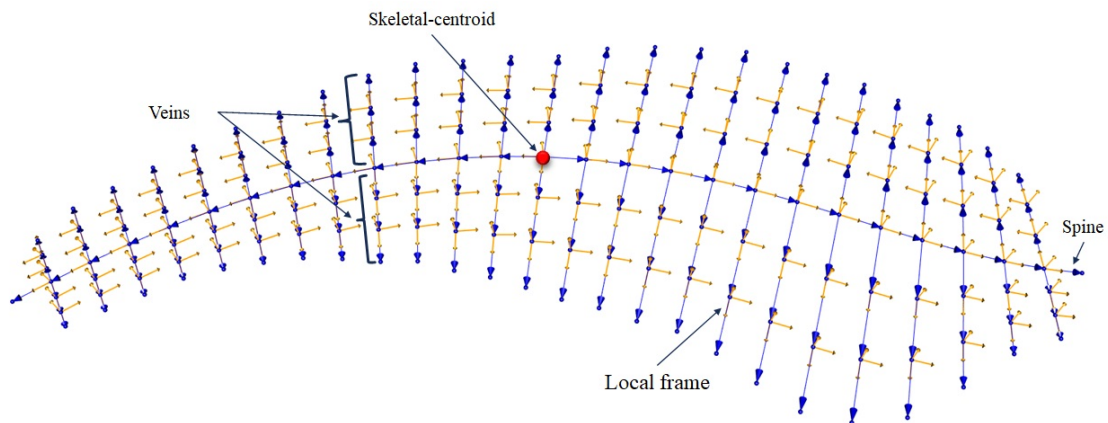


Figure 2: Local parametrization of a hippocampus (Taheri & Schulz (2022).)

Need to learn some basic stuff first!

Before we can start to apply simulation based statistical modelling we need to know something about issues like:

- Random variables
- Probability distributions
- Expectation and variance
- Estimation and limit laws
- How to simulate random variables